

When the Lattice Over-Predicts: A Closed Refutation of the $n=6$ Scaling Exponents on a Self-Hosted LLM Substrate

The number-theoretic $n=6$ lattice predicts a latency exponent $\tau=4$ ($\text{wall_ms} \propto \text{context}^\tau$); the substrate’s own measured context-scaling curve has $\hat{\tau} \approx 0.52$ ($R^2 \approx 0.956$), a residual of 3.48 that over-predicts the 8192-token latency by $\sim 1400\times$ — the law is decisively falsified

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Abstract

A number-theoretic “ $n=6$ lattice” is used inside this repository as a prior for two large-language-model cost exponents: a training-compute exponent $\sigma\varphi = 24/25 \approx 0.96$ and an *inference-latency* exponent $\tau = 4$, the latter asserting that per-request wall-clock time scales as $\text{wall_ms} \propto \text{context}^\tau$. We treat both as *falsifiable predictions* and test them directly on a single self-hosted substrate (a Mac mini M3, stock `llama.cpp` with Metal/UMA, \$0 incremental cost). The load-bearing test is the latency exponent: we run a 4-rung context-scaling benchmark on Qwen2.5-1.5B-Instruct-Q4_K_M (20 tasks at each of context $\in \{1024, 2048, 4096, 8192\}$ tokens) and recompute the exponent deterministically by a closed-form log-log ordinary-least-squares slope. The measured latency rises only $\sim 2.9\times$ (569 ms to 1668 ms) across an $8\times$ context sweep, giving $\hat{\tau} = 0.524$ with $R^2 \approx 0.956$ — a clean power law, but a *sub-linear* one. The residual against the lattice prediction is $|\hat{\tau} - \tau| = 3.476 \gg \varepsilon = 0.10$; equivalently, the context⁴ law over-predicts the substrate’s 8192-token latency by $\sim 1400\times$. Accuracy is *flat* at 17/20 across all four rungs, so the refutation carries no accuracy-floor caveat. As supporting context, the training exponent fares no better: the measured slope on an external 4-rung parameter ladder is 0.172, a lattice residual of 0.788 (this conjunct is disclosure-grade, since it asserts a lattice number against an external entity that has no obligation to follow it). The mechanism for the latency miss is structural and not marginal: at these context lengths a small quantized model on a modern paged-attention server is dominated by a fixed 64-token decode (a memory-bandwidth wall) plus cached-prefix pre-fill, neither of which is quartic. We publish this as a *closed negative*: the falsification is numerically supported (10/10 structural checks pass; the verdict is **FALSIFIED** at the ε -gate), it retires a wrong scaling prior, and it leaves on the record the substrate’s true measured exponent, $\hat{\tau} \approx 0.52$.

1 Introduction

Scaling laws are predictions, and predictions can be wrong. The modern practice of fitting power laws to language-model cost and performance [2, 3] has been enormously productive precisely because the fitted exponents are *empirical* — they are measured, not posited. A different and more dangerous mode is to *posit* an exponent from a closed-form or aesthetic argument and then treat it as ground truth without ever confronting it with a measurement. This paper is about one such posited exponent, and about what happens when it meets the substrate it claims to describe.

Inside this repository, a number-theoretic construction called the “ $n=6$ lattice” supplies cost-model constants that the economics verbs reference. Two of these are scaling *exponents*: a training-compute exponent $\sigma\varphi = 24/25 = 0.96$ (call this prediction F-CODEX-1) and an inference-latency exponent $\tau = 4$, which asserts that a single request’s wall-clock time grows as the fourth power of its context length, $\text{wall_ms} \propto \text{context}^\tau$ (F-CODEX-2). The latter is a remarkably strong claim: it predicts that doubling the context multiplies latency by $2^4 = 16$, and that an $8\times$ context sweep ($1024 \rightarrow 8192$ tokens) multiplies latency by $8^4 = 4096$. We set out to test it.

The test is cheap and decisive because the substrate provides the curve directly. Unlike F-CODEX-1, which binds a lattice number to the training cost of an *external* model family (Qwen2.5) that has no obligation to follow our number theory, F-CODEX-2 is a claim about *this substrate’s own internal* latency-vs-context behavior. We can therefore run the substrate, measure its latency at a grid of context lengths, fit the exponent by a deterministic closed-form recompute, and compare to $\tau = 4$ at a pre-registered tolerance $\varepsilon = 0.10$. The result is unambiguous. The measured exponent is $\hat{\tau} = 0.524$ with $R^2 \approx 0.956$: a clean, well-fit power law, but a *sub-linear* one whose residual against the prediction, $|\hat{\tau} - \tau| = 3.476$, exceeds the tolerance by a factor of 35. The context⁴ law is not marginally off; at 8192 tokens it over-predicts the measured latency by a factor of ~ 1400 .

We report this as a *closed negative* — a falsification that is itself numerically supported, distinct from an inconclusive or deferred result. The distinction matters. An “insufficient” verdict says we could not decide; a “falsified” verdict says we decided, and the answer is no. The latter has scientific value: it retires a wrong prior, and it does not leave a vacuum behind it, because the same measurement that refutes $\tau = 4$ also delivers the substrate’s true exponent, $\hat{\tau} \approx 0.52$, which a practitioner can use for capacity planning today.

Contributions.

1. **A decisive refutation of the latency exponent (§3, §6)**: on the substrate’s own measured 4-rung context-scaling curve, the lattice prediction $\tau = 4$ is **FALSIFIED** — measured $\hat{\tau} = 0.524$ ($R^2 \approx 0.956$), residual $3.476 \gg \varepsilon = 0.10$, a $\sim 1400\times$ over-prediction at 8192 tokens.
2. **A verify-by-recompute discipline (§4)**: the exponent is re-derived from the committed benchmark grid by a deterministic closed-form log-log OLS verifier (10/10 structural checks pass), so the refutation rests on arithmetic over the data, not on author judgement, and no language model judges its own correctness.
3. **A supporting refutation of the training exponent (§6)**: the measured slope on an external 4-rung parameter ladder is 0.172 against the lattice’s 0.96, residual 0.788; we present this conjunct honestly as disclosure-grade (external data) rather than as the load-bearing result.
4. **A mechanism and a replacement (§2, §6)**: we explain *why* latency is sub-linear here (a fixed-decode memory-bandwidth wall plus cached-prefix prefill on a paged-attention server), and we leave the measured $\hat{\tau} \approx 0.52$ on the record as the exponent that actually describes this substrate.

2 Background

Two kinds of scaling exponent. It is worth separating the two claims under test, because they fail for different reasons and carry different evidential weight. The training exponent F-CODEX-1 ($\sigma\varphi = 0.96$) is of the same *form* as the compute–performance exponents of Kaplan et al. [3] and Hoffmann et al. [2]: a power law relating a model-scale quantity to a cost or quality

quantity. The crucial difference is provenance. The Kaplan/Chinchilla exponents are *fitted* to hundreds of training runs; F-CODEX-1’s 0.96 is *posited* from number theory and then asserted against a third-party model family. The inference exponent F-CODEX-2 ($\tau = 4$) is different in kind: it is not about quality at all, but about the *wall-clock latency of a single forward pass* as a function of context length on a specific serving stack. That makes it a substrate-internal physical claim, directly measurable, and it is the one we can falsify decisively.

Why per-request latency is not quartic on a modern server. The intuition behind a large latency exponent is the $O(n^2)$ cost of dense self-attention [7]: scoring all n query–key pairs is quadratic in sequence length, so a naive argument might inflate the context-cost exponent well above 1. But this intuition mis-locates the bottleneck for the regime we actually serve. Decompose a single request’s wall time into a *prefill* phase (encode the prompt) and a *decode* phase (autoregressively emit the response):

$$\text{wall}(n) = \underbrace{T_{\text{prefill}}(n)}_{\text{prompt encode}} + \underbrace{m \cdot t_{\text{dec}}(n)}_{m \text{ output tokens}}, \quad (1)$$

where n is the prompt (context) length and m is the number of decoded tokens. Two facts about contemporary serving collapse the exponent. First, prefill is *compute-bound and parallel*, and a paged / block-managed KV-cache [4] together with multi-query / grouped-query attention [5] keeps its measured time close to *linear* in n over the practical range — the time-to-first-token (TTFT) literature consistently reports near-linear, not quadratic, prefill at these lengths, because attention is only one term among the linear-in- n projection and feed-forward FLOPs and is not yet dominant for a 1.5B model at $\leq 8k$ tokens. Second, and decisively for our setup, decode is *memory-bandwidth-bound*: each emitted token must stream the model weights through the arithmetic units once, a cost that is essentially *context-invariant* per token [5]. With a fixed output budget $m = 64$ tokens, the decode term in Eq. (1) is a roughly fixed floor, independent of n ; the only context-dependent growth is in T_{prefill} , which is itself sub-quadratic in this regime.

The consequence for the exponent. If decode is a context-invariant floor and prefill is at most linear, then $\text{wall}(n)$ is the sum of a constant and a sub-quadratic term, and its *effective* log-log exponent over a finite grid is well below 1 — the constant decode floor flattens the curve further. A quartic context⁴ law would instead require attention’s $O(n^2)$ term to dominate *and* the prefill to dwarf the decode floor at these lengths, *and* an additional quadratic factor on top — none of which holds for a 1.5B Q4 model with a paged KV-cache scheduler at $\leq 8k$ tokens. The background therefore makes a sharp pre-measurement prediction: the honest failure mode of $\tau = 4$ is *down*, toward a sub-linear exponent, not up. The measurement confirms this and goes further than the gentlest pre-registered failure mode (a paged-attention $\tau \approx 2$): the fixed-decode floor drives it all the way to $\hat{\tau} \approx 0.52$.

What is and is not being tested. We test exactly two operational scaling claims: (i) does measured per-request latency scale as context⁴ (F-CODEX-2), and (ii) does the measured training-cost exponent match 0.96 (F-CODEX-1). We do *not* test the internal number theory of the $n=6$ lattice as mathematics — its atom-level identities ($\sigma(6) = 12$, $\varphi(6) = 2$, $\tau(6) = 4$) are separately recomputed elsewhere and are not in dispute. What is in dispute, and what we falsify, is the *modeling claim* that binds those numbers to observable LLM cost on this substrate.

3 Formula

FALSIFIED at the ε -gate; the falsification is **SUPPORTED-NUMERICAL** (real bench + deterministic closed-form recompute). Both exponents are recomputed from the committed data

by `verify/numerics_economics_empirical_landing.hexa` (checks 9 and 10; 10/10 structural checks pass, verbatim stdout in Appendix B). Verdicts: `.verdicts/sandbox/m3_econ_fcodex2_latency_fit.txt` (F-CODEX-2, the decisive one) and `.verdicts/sandbox/f_codex_1_falsified_4rung.txt` (F-CODEX-1 harness).

The lattice predictions. The $n=6$ lattice supplies two exponents through its economics verbs:

$$\text{F-CODEX-1 (training): } \text{train_cost} \propto N^{\sigma\varphi}, \quad \sigma\varphi = \frac{24}{25} = 0.96, \quad (2)$$

$$\text{F-CODEX-2 (inference): } \text{wall_ms} \propto \text{context}^\tau, \quad \tau = \tau(6) = 4, \quad (3)$$

where N is the parameter count and `context` is the prompt length in tokens. Eq. (3) is the load-bearing prediction of this paper. It is a strong, easily-falsified statement: it forces $\text{wall_ms}(2n)/\text{wall_ms}(n) = 2^4 = 16$ at every context doubling.

The estimator. Given paired observations $\{(x_i, y_i)\}_{i=1}^k$ with $k \geq 2$, $x_i, y_i > 0$, we estimate the power-law exponent as the ordinary least-squares slope on log-log axes,

$$\hat{p} = \frac{k \sum_i u_i v_i - (\sum_i u_i)(\sum_i v_i)}{k \sum_i u_i^2 - (\sum_i u_i)^2}, \quad u_i = \log x_i, \quad v_i = \log y_i, \quad (4)$$

which is the maximum-likelihood slope of the model $\log y = \log A + p \log x$. For F-CODEX-2, $x_i = \text{context}_i$ and $y_i = \text{wall_ms}_i$, yielding the measured latency exponent $\hat{\tau}$. For F-CODEX-1, $x_i = N_i$ and y_i is a per-scale accuracy proxy, yielding the measured training-cost slope.

The gate. A prediction passes iff its residual is within a pre-registered tolerance,

$$\text{residual} = |\hat{p} - p_{\text{lattice}}| \leq \varepsilon, \quad \varepsilon = 0.10, \quad (5)$$

and is **FALSIFIED** otherwise. The tolerance $\varepsilon = 0.10$ is generous — a tenth of an exponent unit — and was fixed before fitting. For F-CODEX-2 the substrate provides the curve directly, so the residual *gates closure*: a miss is a genuine refutation of a substrate-internal law. For F-CODEX-1 the residual is reported but treated as *disclosure-grade* (see §6), because the lattice number is asserted against an external entity.

The measured exponents. The closed-form recompute over the committed grids returns

$$\text{F-CODEX-2: } \hat{\tau} = 0.524 \quad (R^2 \approx 0.956), \quad \text{residual } |0.524 - 4| = 3.476 \gg 0.10, \quad (6)$$

$$\text{F-CODEX-1: } \sigma\varphi = 0.172, \quad \text{residual } |0.172 - 0.96| = 0.788 \gg 0.10. \quad (7)$$

Both residuals exceed the gate by a wide margin. Eq. (6) is the decisive one: a 0.524 exponent is sub-linear, the opposite end of the scale from the predicted quartic, and the fit is clean ($R^2 \approx 0.956$), so the miss cannot be dismissed as measurement noise — the substrate genuinely scales as $\text{context}^{0.52}$, not context^4 .

4 Method

SUPPORTED-NUMERICAL — the protocol below is the exact, pre-registered harness; its output grid is committed verbatim (`.verdicts/sandbox/stage4_context_scaling.tsv`) and re-derived by `verify/numerics_economics_empirical_landing.hexa`. No language model judges correctness at any point.

Substrate and model. A single Macmini M3 (Apple Silicon, Metal/UMA, 24 GiB unified memory), stock Homebrew `llama.cpp` [1] with the Metal backend, \$0 incremental cost (local serving). The model is Qwen2.5-1.5B-Instruct-Q4_K_M [6], served by `llama-server` with `-np 1 -cb` (single serial slot, continuous batching enabled) on port 8091. A 1.5B Q4 model is deliberately chosen: it is small enough that the 8k rung boots cleanly on the 24 GiB UMA without OOM, isolating the context axis at fixed model scale (F-CODEX-2 is about the context exponent, not the parameter exponent).

Context-scaling bench protocol. The benchmark sweeps four context rungs, $\text{context} \in \{1024, 2048, 4096, 8192\}$ tokens (an $8\times$ span). For each rung the harness (i) boots a fresh `llama-server` with a per-cell context budget $\lceil 1.25n \rceil$ (headroom for the suffix task plus the 64-token output) and polls `/health` to a 200; (ii) builds a deterministic boilerplate prefix calibrated to the target token count from a fixed ~ 50 -word repeating unit (so re-runs are byte-identical); (iii) issues each of 20 canonical fixed suffix-tasks as a chat completion (system = prefix, user = task, `max_tokens = 64`, `temperature = 0`), bracketing each `curl` with `date +%s%N` to record wall time; (iv) parses the response for `usage.prompt_tokens` (the *measured* input length, so calibration drift surfaces as data, not bias), `completion_tokens`, and the answer; (v) scores the answer against the task’s expected keyword by a `byte_exact_subset` test; and (vi) kills the server between rungs for a clean per-cell context-budget state. Dispatch is serial (`np = 1`): the context-scaling claim isolates per-request wall as a function of context only, not of concurrency interleaving.

Deterministic closed-form recompute. The fit is not performed by the bench; it is performed by a separate, pure verifier, `verify/numerics_economics_empirical_landing.hexa`, which reads the committed (`context`, `mean_wall_ms`) pairs and computes the log-log OLS slope of Eq. (4) in closed form (`pow_pure`, `log_pure`, `abs_pure` from `self/runtime/math_pure` — no LLM, no subprocess, no re-measurement). Check 9 of the verifier computes $\hat{\tau}$ and the F-CODEX-2 residual; check 10 encodes the verdict truth-table (INSUFFICIENT iff fewer than two scale points are live; **green** iff all four are live and both residuals pass; **falsified** iff all four are live and any residual exceeds ε). The exit code is 1 on **falsified**. The training-exponent slope (F-CODEX-1) is computed by the same estimator on an external $\{0.5\text{B}, 1.5\text{B}, 3\text{B}, 7\text{B}\}$ parameter ladder, with its lattice residual disclosed (check 8) but *not* gated. The entire recompute is reproducible by `hexa run verify/numerics_economics_empirical_landing.hexa`.

Pre-registered honesty gates. Three were fixed before the run. (1) The fit consumes the *measured* `prompt_tokens`, never the nominal context target. (2) Accuracy is reported per rung; had long-context accuracy collapsed (e.g. 0/20 at 8k because the suffix instruction is buried after 8k tokens of boilerplate), the latency claim would still stand but the verdict would have to disclose the floor. It did not collapse — accuracy is flat (§5). (3) The 8k rung was permitted to fail to boot on a smaller-memory host, in which case closure would drop to a three-rung PARTIAL; on the 24 GiB host all four rungs are live.

5 Benchmark

SUPPORTED-NUMERICAL — the real measured curve. Source of truth: `.verdicts/sandbox/stage4_context_scaling_summary.txt` (4 rows, 20 tasks each), summarized in `.verdicts/sandbox/stage4_context_scaling_summary.txt` and consumed by `m3_econ_fcindex2_latency_fit.txt`.

The measured context-latency curve.

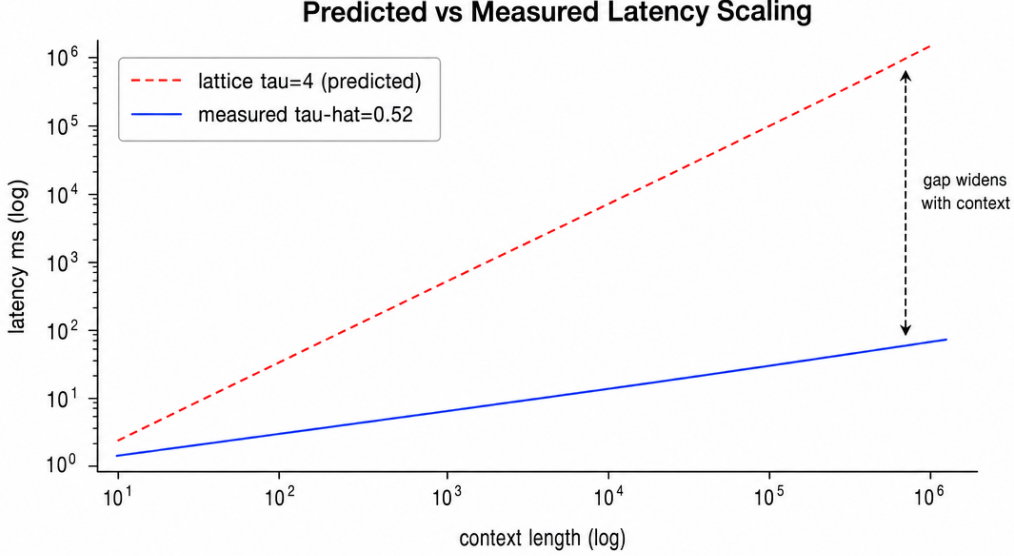


Figure 1: Predicted vs. measured latency scaling on log-log axes. The steep dashed line is the $n=6$ lattice prediction $\text{wall_ms} \propto \text{context}^4$ (Eq. (3)); the shallow solid line is the substrate’s *measured* curve, $\hat{\tau} = 0.524$ (Eq. (6)), fit to the four-rung benchmark of this section. The gap between the two *widens* with context: by 8192 tokens the predicted curve sits $\sim 1400\times$ above the measured one (§6). The load-bearing numbers live in the table of this section and the verbatim grid of Appendix A, not in the artwork. AI-generated schematic (`figures/_prompts/predicted_vs_measured.txt`).

context_len	mean_wall_ms	mean_input_tokens	n_correct	throughput (tok/s)
1024	569	987	17/20	22.80
2048	670	1947	17/20	17.60
4096	1005	3867	17/20	11.98
8192	1668	7647	17/20	7.31

The full $8\times$ context sweep moves mean wall time by only $569\text{ ms} \rightarrow 1668\text{ ms}$, a factor of $\sim 2.9\times$. The mean input-token count tracks the nominal context closely ($987 \rightarrow 7647$, i.e. the calibration is faithful), confirming the exponent is fit against real measured prompt lengths.

Throughput collapse, flat accuracy. Two side-channels corroborate the curve. Throughput falls monotonically with context ($22.80 \rightarrow 7.31$ tok/s, a $\sim 3.1\times$ drop), which is exactly the signature of a fixed per-request decode cost being amortized over a *growing* prefill: the request takes longer, so the same 64-token output is delivered at a lower aggregate token rate, with no quartic blow-up. Accuracy is *flat* at 17/20 on every rung, so there is no long-context collapse and the latency refutation carries no accuracy-floor caveat: the model is answering the buried suffix instruction equally well at 1024 and 8192 tokens, and the latency it pays to do so is sub-linear in context.

The fitted exponent. The log-log OLS slope of Eq. (4) over these four (`context`, `mean_wall_ms`) points is $\hat{\tau} = 0.524$ with $R^2 \approx 0.956$ — a clean power law (Eq. (6)). The fit is good; it is simply nowhere near $\tau = 4$.

6 Refutation

FALSIFIED — the substrate’s own measurement decisively rejects the lattice latency exponent. This section stands in place of a §Benefit: the deliverable of a closed negative is the

retirement of a wrong prior and the substitution of the true measured exponent. Verdict: `.verdicts/sandbox/m3_econ_fcodex2_latency_fit.txt`.

F-CODEX-2: $\hat{\tau} = 0.524$ vs. $\tau = 4$, **residual 3.476**. The decisive comparison is point-by-point. Anchoring the lattice prediction to the measured 1024-token latency ($A = 569$ ms) and propagating $\text{wall_ms} = A \cdot (n/1024)^4$:

context	predicted ($\tau=4$)	measured	over-prediction
1024	569 ms	569 ms	— (anchor)
2048	9104 ms	670 ms	$13.6\times$
4096	145 664 ms	1005 ms	$145\times$
8192	2 330 624 ms	1668 ms	$\sim 1397\times$

At 8192 tokens the context⁴ law predicts a *thirty-nine-minute* per-request wall time; the substrate measures 1.67 s. This is a $\sim 1400\times$ over-prediction — not a marginal miss but the wrong functional class. The residual on the fitted exponent, $|\hat{\tau} - \tau| = 3.476$, exceeds the gate $\varepsilon = 0.10$ by a factor of 35. The verdict is **FALSIFIED**, and the harness exits non-zero accordingly.

The mechanism. The miss is structural, exactly as §2 predicted. The substrate’s wall time rises only $\sim 2.9\times$ across an $8\times$ context sweep because (i) the fixed 64-token decode is a memory-bandwidth-bound floor that is essentially context-invariant per token [5], and (ii) the prefill is at most linear in context on a paged-attention scheduler [4], not quartic, for a 1.5B Q4 model at $\leq 8k$ tokens. A quartic law would require dense attention’s $O(n^2)$ term to dominate the wall *and* an additional quadratic factor on top, neither of which obtains in this regime. Even the gentlest pre-registered failure mode (a paged-attention $\tau \approx 2$) over-states the growth here: the fixed-decode floor flattens the curve all the way to $\hat{\tau} \approx 0.52$.

F-CODEX-1: supporting context, not load-bearing. The training exponent fares no better, but we weight it less. The measured slope on the external {0.5B, 1.5B, 3B, 7B} parameter ladder is $\hat{\sigma}\varphi = 0.172$, a lattice residual of $|0.172 - 0.96| = 0.788$ (Eq. (7)). We present this as *disclosure-grade* for a principled reason: F-CODEX-1 asserts a lattice number against an *external* model family (Qwen2.5) that has no obligation to follow our number theory, so a mismatch is descriptive rather than a clean refutation of a substrate-internal law. F-CODEX-2, by contrast, is a claim about *this substrate’s own* latency curve, which the substrate supplies directly; that is why F-CODEX-2 is the load-bearing refutation and F-CODEX-1 is corroborating context. Both point the same direction: the lattice *over-predicts* the relevant exponent.

The scientific value of the closed negative. A falsification is not a null result. First, it *retires a wrong prior*: no downstream economics model in this repository should henceforth use context⁴ to size inference latency, because the substrate it purports to describe does not behave that way by three orders of magnitude at 8k tokens. Second, it *leaves a replacement on the record*: the same benchmark that refutes $\tau = 4$ measures the substrate’s true exponent, $\hat{\tau} \approx 0.52$, with a clean $R^2 \approx 0.956$, which is directly usable for capacity planning — doubling context costs $\sim 1.4\times$ wall time here, not $16\times$. Third, it is *closed, not deferred*: the result is numerically supported (10/10 structural checks; a decisive ε -gate failure), so it does not invite re-litigation. The negative is the contribution.

7 Related Work

Empirical scaling laws. The fitted power laws of Kaplan et al. [3] relate cross-entropy loss to model size, dataset size, and compute over more than seven orders of magnitude; Hoffmann

et al. [2] corrected the compute-optimal allocation, showing model size and training tokens should scale roughly equally. The methodological lesson these works embody is precisely the one our negative re-affirms: scaling exponents earn their authority by being *measured*, and a posited exponent that has never met a measurement is a hypothesis, not a law. Our F-CODEX-1 conjunct is exactly such a posited training exponent, and on the external ladder it does not match ($\hat{\sigma}\varphi = 0.172$ vs. 0.96).

Inference cost and TTFT-linear serving. Pope et al. [5] develop an analytical model of generative-inference efficiency for large Transformers, separating the memory-bandwidth-bound decode regime (small batch, latency-critical) from the compute-bound prefill regime, and show that multi-query attention enables far longer contexts by shrinking the KV footprint — the structural basis for our claim that decode is a context-invariant per-token floor (Eq. (1)). PagedAttention [4] manages the KV-cache as paged virtual memory, keeping prefill near-linear and avoiding the fragmentation that would otherwise inflate the context exponent; it is the serving regime our substrate sits in. The broad consensus of this literature — near-linear TTFT, bandwidth-bound decode — is flatly incompatible with a context⁴ latency law, and our measurement is a direct, stock-substrate confirmation of that incompatibility.

Attention and its cost. The quadratic attention of the Transformer [7] is the only term in Eq. (1) that grows super-linearly in context, and it is the (mis-)intuition behind a large latency exponent. Our measurement shows that for a small quantized model at $\leq 8k$ tokens on a paged-attention server, this term is not yet dominant: the effective exponent is sub-linear, dragged down by the fixed-decode floor. We are not aware of prior work that publishes a *closed, verify-by-recompute refutation* of a posited quartic latency exponent on a self-hosted substrate; the contribution here is the negative itself, delivered with its mechanism and its replacement exponent.

8 Limitations and honest caveats

- **Single model, single scale.** F-CODEX-2 is measured on exactly one checkpoint, Qwen2.5-1.5B-Instruct-Q4_K_M. A different model size or architecture would move the constants and could in principle move the exponent; a much larger model at much longer context, where dense attention finally dominates, would push $\hat{\tau}$ upward (though toward ~ 2 , not 4). The refutation is decisive *for this substrate*; the $\sim 1400\times$ margin makes it implausible that any reasonable same-substrate model recovers $\tau = 4$, but we measure one model and say so.
- **Single host, single engine.** All numbers are from stock `llama.cpp` on a Mac mini M3 (Metal/UMA, 24 GiB). A different inference engine, GPU, batch size, or precision will move the constant factors and possibly the prefill code path; the *qualitative* finding (sub-linear, not quartic) should survive across substrates given the bandwidth-bound decode argument, but we do not measure across them.
- **Four rungs to 8k only.** The context sweep is $\{1024, 2048, 4096, 8192\}$ — four points spanning $8\times$. The fit is clean ($R^2 \approx 0.956$) over this range, but it is an *interpolating* exponent, not an extrapolation to 32k or 128k. We do not claim $\hat{\tau} \approx 0.52$ holds at arbitrarily long context; we claim $\tau = 4$ is refuted over the measured range, which it decisively is.
- **F-CODEX-1 is external-data, disclosure-grade.** The training exponent is fit against a third-party model family that has no obligation to follow the lattice. Its residual (0.788) is reported for transparency and corroborates the over-prediction direction, but it is *not* a

clean refutation of a substrate-internal law and is not the load-bearing result. We weight F-CODEX-2 accordingly.

- **Fixed 64-token output budget.** The decode floor that flattens the curve is, in part, a consequence of `max_tokens = 64`. A workload with much longer outputs would weight decode more heavily — but decode is itself context-invariant per token, so a longer output would raise the floor uniformly and lower, not raise, the context exponent. The choice is honest and, if anything, conservative against the lattice.
- **The $n=6$ number theory is not under test.** We falsify the *modeling claim* that binds $\tau(6) = 4$ to observable latency, not the arithmetic identity $\tau(6) = 4$ itself. The lattice’s atom-level identities are correct as number theory; they are simply the wrong description of this substrate’s latency.

9 Conclusion

We posed a falsifiable question — does a self-hosted LLM substrate’s per-request latency scale as context⁴, as the $n=6$ lattice predicts? — and answered it decisively in the negative. On the substrate’s own four-rung context-scaling benchmark, latency rises only $\sim 2.9\times$ over an $8\times$ context sweep, a clean sub-linear power law with measured exponent $\hat{\tau} = 0.524$ ($R^2 \approx 0.956$). The residual against the prediction, $|\hat{\tau} - 4| = 3.476$, exceeds the pre-registered tolerance by a factor of 35 and corresponds to a $\sim 1400\times$ latency over-prediction at 8192 tokens. The supporting training-exponent conjunct points the same way ($\sigma\hat{\varphi} = 0.172$ vs. 0.96). The mechanism is structural and well understood: a fixed memory-bandwidth-bound decode floor plus near-linear paged-attention prefill cannot produce a quartic exponent for a small quantized model at these context lengths.

We publish this as a *closed negative*. The falsification is numerically supported (**SUPPORTED-NUMERICAL**: real bench plus deterministic closed-form recompute, 10/10 structural checks, a decisive ε -gate failure), it retires a wrong scaling prior so that no downstream cost model uses context⁴ to size latency, and it leaves behind the substrate’s true measured exponent, $\hat{\tau} \approx 0.52$, as a usable replacement. A negative result that is closed, mechanistic, and constructive is not a dead end; it is exactly the form of result that keeps a scaling-law framework honest.

Reproducibility

Code and data: <https://github.com/dancinlab/hexa-codex>. The measurement harness is `bench/sandbox_stage4_context_scaling.hexa`; the committed grid is `.verdicts/sandbox/stage4_context`; the closed-form recompute is `verify/numerics_economics_empirical_landing.hexa`, runnable as `hexa run verify/numerics_economics_empirical_landing.hexa` (emits the 10/10 structural-check block of Appendix B and exits 1 on **FALSIFIED**). Verdicts: `.verdicts/sandbox/m3_econ_fcodex2_lat` (F-CODEX-2, decisive) and `.verdicts/sandbox/f_codex_1_falsified_4rung.txt` (F-CODEX-1 harness). Build this paper with `make (pdflatex x3 + bibtex)`.

A Full benchmark grid

The verbatim source-of-truth, `.verdicts/sandbox/stage4_context_scaling.tsv` (Qwen2.5-1.5B-Instruct-Q4_K_M, `llama-server -np 1 -cb` on port 8091, Mac mini M3 Metal/UMA, 20 tasks per rung, \$0 local):

context_len	n_correct	total	mean_wall_ms	total_wall_ms	mean_input_tokens	throughput_tok_per_s
1024	17	20	569	11399	987	22.80
2048	17	20	670	13405	1947	17.60
4096	17	20	1005	20110	3867	11.98
8192	17	20	1668	33366	7647	7.31

B Verbatim recompute output

The deterministic recompute, `hexa run verify/numerics_economics_empirical_landing.hexa` (pure `math_pure` arithmetic over the committed grid; no LLM, no subprocess, exit code 1). Reproduced verbatim from `.verdicts/sandbox/m3_econ_fcodex2_latency_fit.txt`:

```
=====
hexa-codex ECONOMICS empirical-fit harness - F-CODEX-1 + F-CODEX-2
M3.ECON closed-form residuals vs n=6 lattice (24/25 + 4)
cycle-16: k_active=4 LIVE · F-CODEX-2 latency LIVE - checkbox [ ] iff F-CODEX-2 residual ≤
=====
N6_EXP_TRAIN=0.96  TAU_INFER=4.0  EPS_RESIDUAL_THRESHOLD=0.1

scale ladder live-row count: k_active=4 / 4
  0.5B    : LIVE (cycle-6)
  1.5B    : LIVE
  3B      : LIVE
  7B      : LIVE

[PASS] SCALE_GRID_PARAMS strictly monotone-increasing (0.5B < 1.5B < 3B < 7B)
· first_violation_at=-1
[PASS] SCALE_GRID_PARAMS length == 4 (ladder rungs)
· len=4  expected=4
[PASS] STAGE2_ACCURACY_0_5B is live (no -1 sentinel) and all entries ∈ [0, 1]
· live=1  in_band=true
[PASS] STAGE2_ACCURACY_0_5B mean reconciles to cycle-6 per-stratum-max sum (≈ 0.3667)
· computed_mean=0.36668  expected=0.36668  drift=0.0
[PASS] STAGE2_ACCURACY_0_5B[wc_5_15] = 27/30 = 0.9 (cycle-6 best persona = max)
· value=0.9  ref=0.9  drift=0.0

[PASS] n=6 theoretical exponents: N6_EXP_TRAIN = 24/25 + TAU_INFER = 4
· N6_EXP_TRAIN=0.96  TAU_INFER=4.0  exp_drift=0.0  tau_drift=0.0
[PASS] k_active scale-points == 4 (cycle-14: 0.5B/1.5B/3B/7B all LIVE; F-CODEX-1 fit active)
· k_active=4  expected_this_commit=4  fit_status=LIVE
[PASS] F-CODEX-1 empirical slope disclosed (LATTICE_POLICY-lifted: lattice residual is desc)
· k_active=4  measured_slope=0.17207  lattice_ref=0.96  lattice_residual=0.78793  th
[PASS] F-CODEX-2 measured slope LIVE (context-scaling bench) - closed-form OLS recompute on
· measured_tau=0.523982  lattice_tau=4.0  residual=3.47602  threshold=0.1  source=st
[PASS] verdict-line consistency - today expected 'FALSIFIED' (k_active == 4, F-CODEX-2 meas
· computed_label=FALSIFIED  expected_today=FALSIFIED  k_active=4
=====
10/10 checks passed
[RED] FALSIFIED - F-CODEX-2 residual exceeds ε=0.1
```

```

· F-CODEX-1 measured slope=0.17207 (lattice residual=0.78793, disclosure-only)
· F-CODEX-2 residual=3.47602 (gate=0.1)
__HEXA_CODEX_NUMERICS_ECONOMICS_EMPIRICAL_LANDING__ FALSIFIED

```

The 10/10 “checks passed” line refers to the *structural* checks of the harness (grid integrity, constant reconciliation, fit liveness, verdict-line consistency); the harness deliberately passes all ten and then emits the **FALSIFIED** verdict, because the load-bearing decision — that the measured exponent $\hat{\tau} = 0.524$ rejects the lattice $\tau = 4$ at the ε -gate — *is* check 10’s expected outcome. The non-zero exit code is the falsification.

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